

Optimizing Skill Allocation to Reduce Unemployment Using PLFS 2023–24 Data

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Abstract

This project aims to develop an optimization-based approach to allocate vocational training budgets across Indian states and union territories in a way that helps reduce unemployment. Rather than explicitly modeling the impact of training on unemployment reduction—which is complex and data-intensive—the strategy involves using observable metrics from the Periodic Labour Force Survey (PLFS) and other public datasets to guide how resources should be distributed. The model considers key indicators such as unemployment rates, state population proportions, GDP figures, and rural-urban classifications to prioritize regions for budget allocation.

The optimization model is designed to minimize a weighted sum of unemployment across regions by allocating budget shares proportionally. Constraints ensure that the entire budget is used, allocations are non-negative, and a minimum share is directed to rural-dominant areas.

1 Problem:

Unemployment remains one of the most persistent socio-economic challenges in developing economies like India. While the government has launched numerous employment schemes and skill development initiatives, certain states and demographic groups continue to experience high unemployment rates. A critical factor exacerbating this issue is the mismatch between worker skills and labor market needs.

This project aims to develop a data-driven optimization model for the efficient allocation of vocational training and employment-related resources across Indian states and union territories (UTs), using data from the **Periodic Labour Force Survey (PLFS) 2023–24**. Although vocational training has proven effective in improving employability, limited public budgets necessitate optimal allocation strategies to ensure maximum impact. Our objective is to minimize a weighted measure of unemployment burden while satisfying constraints related to equity and rural support.

1.1 Approach and Methodology

Our solution strategy is built on realistic, data-driven principles, incorporating practical policy considerations. Key steps include:

- **Model Selection:** We chose a linear programming model for interpretability, scalability, and computational efficiency.
- **Decision Variable:** Resource allocation per region (state/UT).
- **Weighting:** Weights inversely proportional to Gross State Domestic Product (GSDP) were assigned to favor economically weaker states.
- **Effectiveness Coefficients:** Region-specific factors k_i model how effectively training programs reduce unemployment.
- **Constraints:** Included full resource allocation, non-negativity, and a minimum rural support constraint to ensure equity across regions.

1.2 Motivation

- The need for efficient public spending amid limited fiscal capacity.
- Emphasis on **evidence-based governance** and **data-driven decision-making**.
- Traditional resource allocation (e.g., equal distribution) fails to capture local disparities and economic realities.
- Ensuring training translates to actual employment gains, not just symbolic program participation.

2 Data Collection and Parameter Derivation

2.1 Overview

The data used in this project comes from the **Periodic Labour Force Survey (PLFS) 2023–24**, conducted by the **Ministry of Statistics and Programme Implementation (MoSPI)**, Government of India. This nationally representative survey provides estimates of employment, unemployment, labour force participation, and skill acquisition across regions and demographic groups. It forms the backbone of our optimization-based allocation model aimed at minimizing unemployment through vocational training interventions.

Approximately 100,000 households are surveyed annually using face-to-face interviews by trained field investigators, ensuring temporal and spatial coverage. This is not a census but a probabilistic sample.

2.2 Parameter Extraction for Optimization

The following parameters are derived from the cleaned and processed data to formulate the optimization problem:

- U_i : State-wise unemployment rates
- P_i : state wise population proportion
- $w_i \propto \frac{1}{\text{GSDP}_i}$: Weights inversely proportional to Gross State Domestic Product, prioritizing economically weaker states
- R_i : Rural share of the population (used for minimum rural allocation constraints in the problem)
- k_i : Region-specific effectiveness coefficients estimating employment gain per unit of training investment.(estimated from previous datasets.)

All variables converted into numeric vectors in R. Missing values are handled through column filtering based on a 25% NA threshold:

$$\text{NA Ratio} = \frac{\text{Number of NA entries}}{\text{Total entries}} \leq 0.25$$

2.3 Reason For this Approach:

This "Systematic data pipelining" ensures that the inputs to the optimization model are clean, consistent, and policy-relevant. Using P.L.F.S data enables us to ground the allocation strategy in real-world labor market dynamics, targeting the highest-impact interventions within the budget constraints.

2.4 Parameter Derivation

Definition of U_i (Unemployment Rate)

U_i represents the **current unemployment rate** for a given state or union territory i , as derived from the PLFS dataset.

$$U_i = \frac{\text{Number of Unemployed Persons in State } i}{\text{Number of Persons in the (Labour Force + Unemployed) in State } i}$$

This is calculated algorithmically using the PLFS person-level data (CPERV1) as follows:

Listing 1: Computing $U - i$ in R from PLFS data

```
cperv1-rel <- cperv1-df %>%
  filter(age >= 15) %>%
  mutate(
    unemployed = cws-status %in% c(81, 82),
    labor-force = cws-status %in% c(11, 12, 21, 31, 41, 42,
    51, 61, 62, 71, 72)
  ) %>%
```

```
group-by( state ) %>%
  summarise(unemployment-rate = sum(unemployed) / sum(labor+unemployed))
```

- ‘cws-status’ refers to Current Weekly Status classification in PLFS. - Codes 81 and 82 indicate unemployment (available and seeking work). - Labor force includes those classified as employed or actively seeking employment

The reason why we are using cperv1 to get unemployment rate is because the plfs 2023-24 contains lots of missing values. And our decision variables and parameters are state/UTs wise, however due to missing value for UR FOR 9 states/UTs, we chose to derive it from cperv1. For accuracy purpose we have taken this step.

Definition of W_i :

Weight calculation based on GSDP(Gross State Domestic Product)

We define the priority weight w_i for each state or UT as:

$$w_i = \left(\frac{1}{\text{GSDP}_i} \right) * 10^6$$

where:

- GSDP_i is the Gross State Domestic Product of region i
- $\frac{1}{\text{GSDP}_i}$ is multiplied by 10^6 for computational ease

This formulation ensures that economically weaker states (with lower GSDP_i) are assigned higher weights w_i , thereby receiving higher priority in the allocation process.

Estimation of k_i Values

To estimate the effectiveness coefficient k_i for each region, we used the observed change in unemployment rate before and after benefit disbursement through the A.B.R.Y (Atmanirbhar Bharat Rojgar Yojana) program, following this formula:

$$k_i = \frac{\Delta U_i}{\text{Benefit}_i}$$

Where:

- $\Delta U_i = U_i^{\text{before}} - U_i^{\text{after}}$ is the change in unemployment rate between FY 2020–21 and 2022–23,
- Benefit_i is the total amount (in crore) disbursed to state i under ABRY.

This formulation assumes that the change in unemployment can be partially attributed to employment-linked benefit disbursement and reflects a state’s response to employment initiatives.

The result is a state-specific k_i , representing estimated **marginal reduction in unemployment rate per unit of benefit received**.

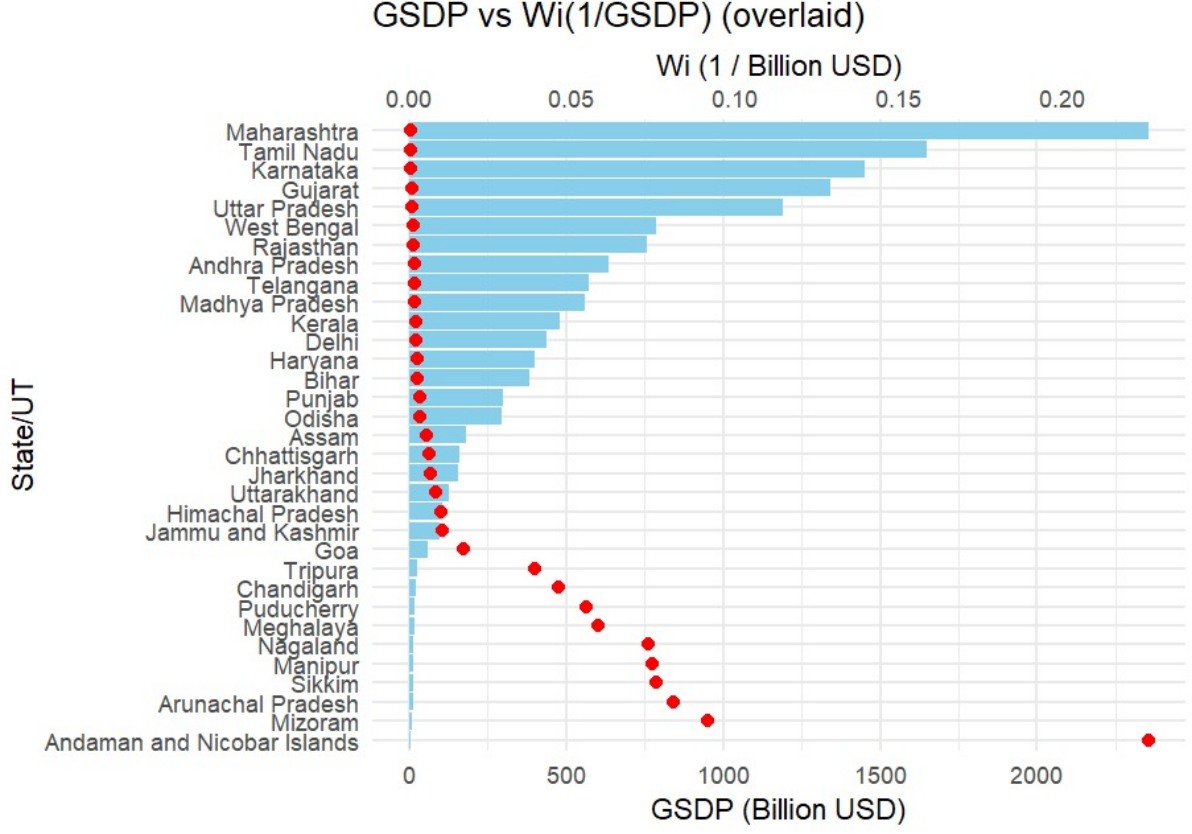


Figure 1: GSDP and their respective weights

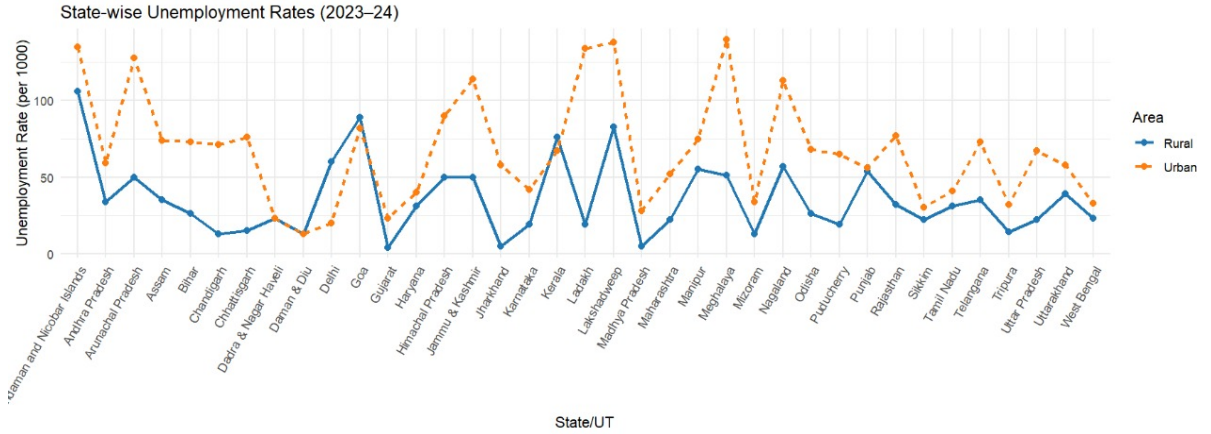


Figure 2: State-Wise unemployment Rate(2023-'24)

3 Objective Function

$$\min z = \sum_{i=1}^n W_i \cdot U_i^{\text{current}} \cdot P_i \cdot (1 - x_i \cdot k_i) \quad (1)$$

Where:

- W_i = weight factor (economic priority, e.g., inverse GDP)

- U_i^{current} = current unemployment rate in region i
- P_i = population proportion of region i
- x_i = budget share allocated to region i
- k_i = impact coefficient of budget on unemployment reduction

4 Mathematical Analysis of the Optimization Model

4.1 Objective Function:

$$\min_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x}) = \sum_{i=1}^n w_i U_i p_i (1 - k_i x_i),$$

where:

- $\mathbf{x} = (x_1, x_2, \dots, x_n)^T$ is the decision variable vector representing allocation shares.
- $w_i \sim 1/\text{GSDP}_i$ is a weight emphasizing poorer regions.
- U_i is the unemployment rate in region i .
- p_i is the population proxy of region i .
- k_i is the effectiveness of allocation x_i in reducing unemployment.

Each term $w_i U_i p_i (1 - k_i x_i)$ is scalar-valued. So $f(\mathbf{x}) \in \mathbb{R}$, and the function maps $\mathbb{R}^n \rightarrow \mathbb{R}$, making it a scalar field.

Unemployment Burden Proxy: $U_i \cdot P_i$

$$U_i \cdot P_i = \left(\frac{\text{Unemployed}_i}{\text{Labor Force}_i + \text{unemployed}} \right) \cdot \left(\frac{\text{Population}_i}{\text{Total Population}} \right)$$

Reflects:

- Size of the state's population (P_i),
- How much of that population is unemployed (U_i).

Core of the objective function:

$$z = \sum_{i=1}^n W_i \cdot U_i \cdot P_i \cdot (1 - x_i \cdot k_i)$$

4.2 Dimensional Consistency

- U_i : dimensionless (rate)
- p_i : number of people
- w_i : dimensionless (derived from inverse GSDP)
- x_i : dimensionless (proportion)

- k_i : effectiveness per unit allocation (dimensionless)

So $w_i U_i p_i (1 - k_i x_i)$ has units of people $\Rightarrow f(\mathbf{x})$ measures total weighted unemployed people remaining.

4.3 Constraints and Feasible Set

Subject to:

$$\sum_{i=1}^n x_i = 1, \quad x_i \geq 0 \forall i, \quad \mathbf{x}^T (\mathbf{p} * \mathbf{R}) \geq \delta,$$

where $\delta = 0.02$ and $\mathbf{R}$ is the rural ratio vector.

The standard n -simplex:

$$\Delta^{n-1} = \left\{ \mathbf{x} \in \mathbb{R}^n \mid x_i \geq 0, \sum_{i=1}^n x_i = 1 \right\},$$

is:

- **Closed**: defined by continuous equality/inequality constraints.
- **Bounded**: $x_i \in [0, 1]$
- Hence, **Compact** (Heine–Borel theorem).

The rural constraint defines a closed half-space. Intersection of convex sets is convex $\Rightarrow$ feasible set is convex and compact.

4.4 Convexity and Existence of Solution

- Objective function is linear in $\mathbf{x} \Rightarrow$ convex.
- Feasible set is compact and convex.
- By *extreme value theorem*, a global minimum exists.

4.5 Algorithmic Path:

- Standard **Linear Program (LP)**.
- Solvable via **Simplex Method** or Interior Point Methods.
- R implementation: **lpSolve** package (revised simplex).

5 Data Required for the Project:

- **Unemployment Rates**: From P.L.F.S
- **Population Proportion**: From R.B.I.'s website
- **Inverse of GDP Figures**: From R.B.I website(W_i)

- **Rural/Urban Split:** P.L.F.S

6 Implementation Considerations:

- **Proxy Use:** Some data (e.g., population) inferred from P.L.F.S samples.
- **Scalability:** Model can adapt to new P.L.F.S releases or with additional indicators.
- **Direct Computation:** Avoids modeling intermediate causal effects.

7 Conclusion, Discussion, and Future Aspect:

7.1 Discussion

This study proposes a linear programming model to optimally allocate vocational training resources across Indian states and UTs, aiming to reduce unemployment. Using P.L.F.S data and computed impact coefficients (k-values), the model identifies regions where training investment yields the highest employment gains.

Findings reveal substantial regional disparities in training effectiveness, supporting the need for region-specific interventions over uniform allocation. States with higher k-values warrant priority to maximize employment gains within fiscal limits.

Nonetheless, the model assumes a linear and measurable relationship between training and employment, which may oversimplify complexities such as informal labor, skill mismatches, and varying policy efficacy.

7.2 Limitations

- Assumes linearity in training impact, potentially limiting real-world accuracy.
- Lacks demographic granularity (e.g., youth proportion in the total population).
- Ignores qualitative aspects like training quality and industry alignment.
- Focuses on a static budget scenario.
- Requires validation through real-world outcomes.

7.3 Future Aspect:

Several directions can extend this work:

- **Temporal Modeling:** Extend the static model into a time-series framework to capture long-term trends and make the model dynamic.
- **Predictive Modeling:** Integrate machine learning (e.g., random forests, gradient boosting) to capture non-linear impacts and enhance adaptive planning.
- **Cost-Benefit Analysis:** Quantify return on investment using training costs and wage outcomes.

- **Dynamic Monitoring:** Develop real-time tracking tools for outcome-based real-location.
- **Policy Simulation:** Model counterfactual scenarios under alternate allocations and constraints.

7.4 References:

- **Reference:** Boyd and Vandenberghe, *Convex Optimization*, Chapter 2.2–2.3; Luenberger and Ye, *Linear and Nonlinear Programming*, Chapter 3.
- <https://github.com/AR3141529/OptimizationProject>(Github Repository)
- **Link 1:** mospi.gov.in/download-reports
- **Link 2:** [microdata.gov.in/PLFS catalog](https://microdata.gov.in/PLFS%20catalog)
- **Link 3:** [dge.gov.in PLFS 2024 Summary \(PDF\)](https://dge.gov.in/PLFS%202024%20Summary%20(PDF))
- **Link 4:** <https://pib.gov.in/PressReleasePage>
- **Link 5:** <https://www.rbi.org.in/scripts/AnnualPublications.aspx?head=Handbook>

7.5 Publications associated with this project:

- IGIDR Working Paper: Employment Dynamics and Employment Guarantee Scheme
- Book: The Global Labour Market by Springer
- NBER Working Paper No. 18859: Youth Unemployment and Vocational Training

7.6 Software and Tools

The analysis is conducted in **R**, using the following packages:

- **dplyr, tidyr** — For Data manipulation
- **readr** — For Data import
- **ggplot2, plotly** — For Visualization
- **lpSolve** — For Linear programming solver

We use data from PLFS 2023-24: CPERV1 (Common Person-Level Data Version I) and CHHV1 (Common Household-Level Data Version I) in CSV format. We also used the LPFR (Labour Force Participation Rate), WPR (Worker Population Ratio), UR(Unemployment Rate) data from plfs 2023-24. We use person-level and household-level CSV datasets from PLFS 2023-24.